THE MINOTAUR'S LABYRINTH

BUILT TO DEFEAT MEMORY. UNDONE BY A THREAD.

STEP ONE

System boundaries & functional unit

SUBJECT

The Labyrinth is modeled as a subterranean masonry confinement and navigation-denial system: large enough to hold one Minotaur, complex enough that an unaided entrant cannot recover the exit.

FUNCTIONAL UNIT

Construction and 30 years of material maintenance for one two-level Labyrinth with a 22,000 m2 plan footprint, designed to confine one Minotaur and defeat unaided navigation.

DENOMINATOR

One labyrinth. 30 years.

Ariadne's thread is quantified separately, not added to the headline.

The 22,000 m2 footprint is borrowed from the official scale of the Palace of Knossos only as a transparent benchmark. It is not an archaeological measurement of the mythical Labyrinth.

Ancient function: Pseudo-Apollodorus, Library 3.1.4 and Epitome 1.8-1.9. Complete formulas and assumptions are in the workbook.

STEP ONE

What the ancient sources actually say

DIRECT LITERARY EVIDENCE

- Daedalus constructed a chamber whose tangled windings perplexed the outward way.
- A person entering could not find the exit because winding turns concealed it.
- The Minotaur was confined and guarded inside the structure.
- Ariadne obtained the escape method from Daedalus.
- Theseus fastened the clue at the door and drew it behind him on the way in and out.

WHAT THE SOURCES DO NOT PROVIDE

- No footprint, levels, wall thickness or passage width
- No bill of materials, construction technology or service life
- No thread length, fiber, mass or reuse history

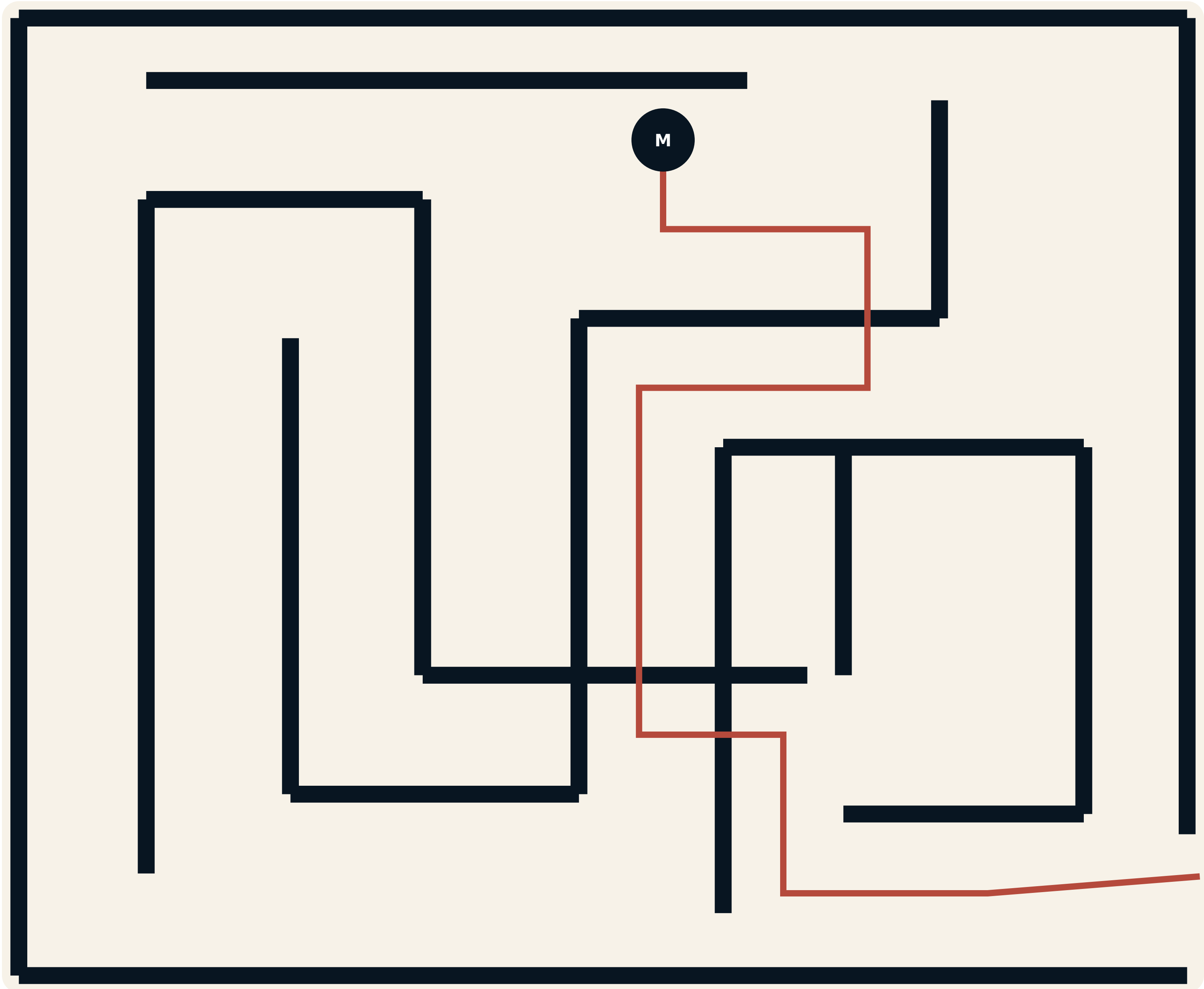
EVIDENCE RULE

Function is ancient. Geometry is reconstructed.

Sources: Pseudo-Apollodorus, Library 3.1.4 and Epitome 1.8-1.9, ToposText; Diodorus Siculus, Library 4.77.4, Theoi.

STEP TWO

A Knossos-scale maze



PLAN FOOTPRINT

22,000 m2

Knossos-scale benchmark

GROSS FLOOR AREA

44,000 m2

Two modeled levels

EXCAVATION

176,000 m3

Disclosed, not carbonized

Official scale benchmark: Hellenic Ministry of Culture, Palace of Knossos, 22,000 m2. The maze icon is explanatory, not archaeological.

STEP TWO

Life-cycle inventory

Worked stone	180,787 t	75.1% GWP
Bricks	8,800 t	12.1% GWP
Wood	1,980 t	4.2% GWP
Metals	88 t	2.1% GWP
Plaster / render	5,133 t	6.3% GWP
Asphalt / bitumen	440 t	0.2% GWP

INITIAL CONSTRUCTION

17.19 kt CO2e

92.4% of the 30-year result

Quantities are formula-driven from editable geometry. DEFRA 2026 Material use factors are transcribed in the workbook with exact rows.

STEP TWO

The product is disorientation

A maze is not defined by floor area alone. Its function emerges from the network of possible movement and the entrant's inability to preserve orientation.

TRAVERSABLE FLOOR

22,000 m2

AVERAGE WIDTH

2.2 m

NETWORK LENGTH

10.0 km

ROUTE TO CENTRE

3.0 km

30% of modeled network

INDICATIVE JUNCTIONS

556

One decision every 18 m

These indicators do not solve the topology. They make the literary requirement auditable: enough route length and repeated decisions to turn memory into the failure mode.

FUNCTIONAL METRIC

Information loss per metre of stone.

Navigation geometry is an engineering assumption: 50% traversable floor, 2.2 m average width, 30% route to centre, 18 m per junction.

STEP TWO

Thirty years of keeping it impossible

The Labyrinth must remain structurally intact, dark, confusing and closed. Humidity and wear turn finishes, wood and waterproofing into recurring material flows.

MAINTENANCE BURDEN

1.41 kt CO2e

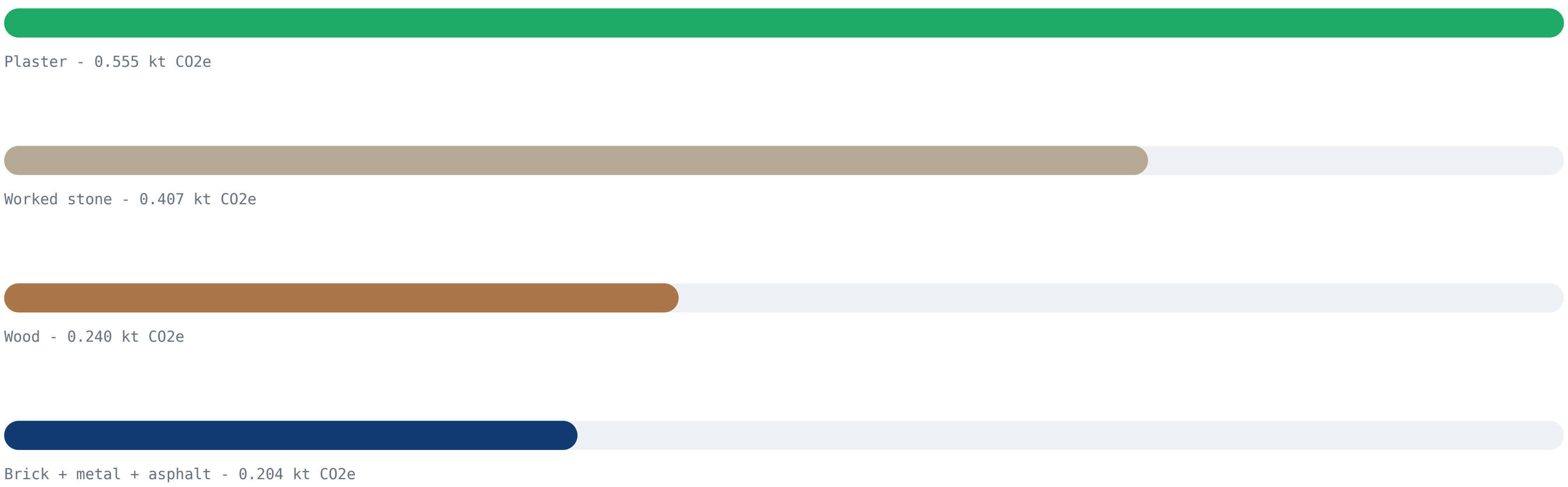
7.6% of lifecycle GWP

LARGEST MAINTENANCE

Plaster

0.555 kt CO2e

THIRTY-YEAR MAINTENANCE BY MATERIAL



MAINTENANCE LESSON

The walls endure. The surfaces keep returning.

Replacement assumptions: stone 0.1%; brick 0.2%; wood 1.5%; metal 0.5%; plaster 3%; asphalt 5% per year.

STEP TWO

The lowest-carbon component wins

Ariadne does not strengthen the building, illuminate it or shorten the route. She changes what Theseus knows about the route he has already travelled.

THREAD MASS

24 kg

3,000 m x 8 g/m

THREAD FOOTPRINT

0.54 t CO2e

Conservative clothing proxy

LABYRINTH / THREAD RATIO

34,700 : 1

The building emits roughly 34,700 times more than the intervention that defeats it.

WHY IT WORKS

- The door becomes an external reference point.
- Every turn is recorded without storing the maze in memory.
- The return path is encoded physically, not reconstructed cognitively.

The maze is a material solution to an information problem.

Thread proxy: 3 km route x 8 g/m = 24 kg. DEFRA 2026 Clothing - primary material production: 22,310 kg CO2e/t.

STEP TWO

What stays outside the maze

PHYSICAL EXCAVATION DISCLOSED

176,000 m3

Ancient excavation energy and labor are not carbonized

EXCLUDED FROM THE HEADLINE

- Lighting, ventilation, food and water inside the Labyrinth
- The fourteen Athenian victims and all consequences of sacrifice
- The Minotaur's food, metabolism and supernatural origin
- Theseus' weapons, travel and the voyage to Crete
- Quarry transport, construction labor and temporary settlement
- Demolition, collapse, abandonment and archaeological excavation

BOUNDARY LESSON

The model is structure + maintenance, not the whole myth.

STEP THREE

Results

Global Warming Potential - one modeled Labyrinth over 30 years

Lifecycle stage / item	kt CO2e	Share
Initial construction	17.190	92.4%
Material maintenance	1.406	7.6%
LABYRINTH TOTAL	**18.595**	**100.0%**
Ariadne's thread	0.000535	separate

PER GROSS FLOOR AREA

422.6 kg CO2e/m2

30-year result / 44,000 m2

PER NETWORK KILOMETRE

1.86 kt CO2e

10 km modeled passage network

WORKED-STONE SHARE

75.1%

Largest lifecycle material

DEFRA COVERAGE

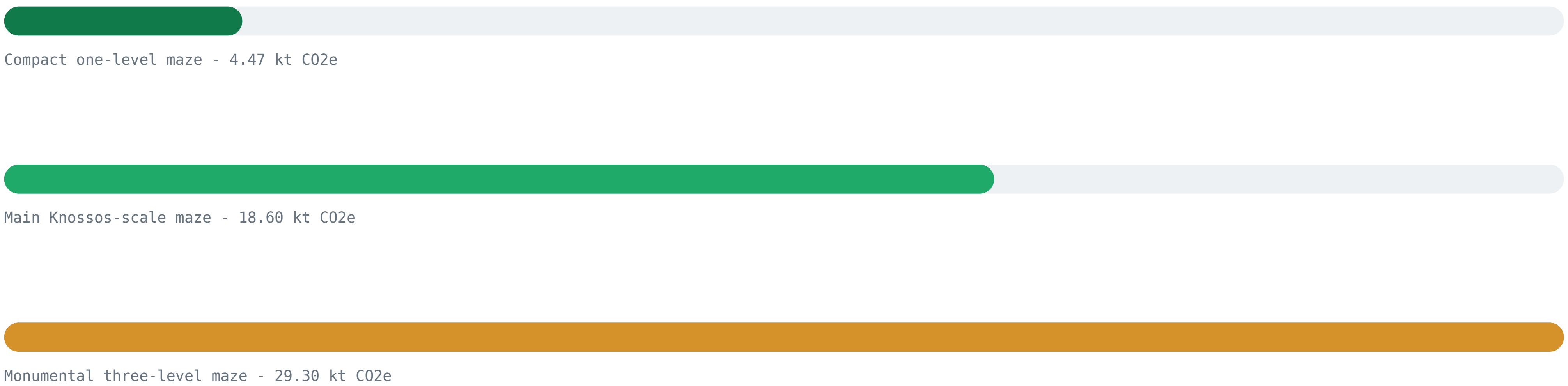
100%

Of quantified main GWP

STEP THREE

Sensitivity: scale and stone

GEOMETRY SCENARIOS



WORKED-STONE CASE

18.60 kt

74.9958 kg CO_2e/t

AGGREGATE PROXY

6.08 kt

7.80307 kg CO_2e/t

The lower factor is shown to expose proxy risk, not to nominate a preferred answer. Unbound aggregate is not a finished load-bearing wall. The largest uncertainty is the material definition, not the fifth decimal place.

SENSITIVITY LESSON

The wall factor can move the answer by 67%.

STEP FOUR

Interpretation & conclusion

The Labyrinth's largest material was not stone.
It was uncertainty.

At Knossos scale, the reconstructed structure contains 180,787 tonnes of worked stone and 10 km of modeled passage network. Its 30-year footprint is 18.6 kt CO2e.

LARGEST MATERIAL

Worked stone

75.1% of lifecycle GWP

WINNING COMPONENT

A thread

0.54 t CO2e, modeled separately

The paradox is architectural: a massive material system is built to create an information deficit, then a 24 kg line defeats it by preserving one bit of state - the way back.

The lowest-carbon component defeats the whole system.

Do not solve an information problem with more material

until you have tested the thread.

Scenario arithmetic is evidence of model transparency - not evidence that the Labyrinth existed.

MYTHS & LEGENDS

Where ancient stories meet measurable emissions

Alberto Carmignani